SAMPLE

Great Observatories: Roman Space Telescope

Wide-Field Cosmic Survey

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Introduction

Somewhere in the queue of great telescopes there is a machine designed to see everything at once. The great observatories of the astronomical tradition have mostly been instruments of depth: narrow-eyed collectors built to stare at one faint object for hours, to go deeper, darker, and further than any eye before them. The subject of this volume is the opposite bet, a telescope built for width, a machine whose whole design is a wager that the next decade's greatest discoveries will come not from drilling a single deep hole in the sky but from surveying it whole, that the universe's biggest questions, the ones written in the behavior of millions of galaxies, need a lens as wide as the questions are broad. This is the wide-field telescope that now stands as the space observatory program's great surveyor, and its story is this book's subject.

The machine's own name carries the other half of its story. The telescope is named for Nancy Grace Roman, the astronomer who, more than any single person, made the idea of space-based astronomy a working program rather than a dream, the founder and first director of the office that planned and sold the first great space telescope and who spent a career arguing that a point of light worth studying was worth a campaign of instruments. Her story and the machine's story are this volume's twin threads: the person who taught the space agencies to build observatories, and the observatory built, generations later, in her name, a mirror of the history as neat as the sky provides.

The reader should know at the outset what kind of book this is, because the subject is unusual among the great observatories: it is a book partly about a promise. The machines this series otherwise celebrates have decades of operations behind them; this telescope's career is largely ahead of it, planned, built, and tested but with its discoveries still in the future, and the honest volume about it is a volume about a design, its science case, its engineering, and the reasons the astronomical community bet a flagship mission on a wide eye. The chapters therefore lean on the spacecraft's heritage, its instruments, and its planned campaigns, and on the mathematics of the sky that no telescope can escape, the dark energy that needs a census, the exoplanets that need a search, the infrared sky that needs a map, all of which the wide eye was designed to deliver.

A word on neighbors: the great infrared observatory already flying has its own volume in this series, and the present book touches its story only at the border, where the two machines' different wagers, depth versus width, explain each other; the history of the earlier space telescopes, the famous one named for the classifier of galaxies and the great infrared successor, is likewise told where it belongs, and here only as heritage. What follows is the story of the wide eye and the woman it honors: the design's logic, the hardware's journey from a spy agency's closet to the space agency's flagship, the science that justified it, and the survey decade it promises, a book about a telescope whose best chapters, by design, have not happened yet.

A paragraph on the machine's address in this series' family, because the reader deserves the map before the tour. The great observatories of this publishing program span the whole spectrum and both of the sky's shapes: the pointed giants of the optical, the infrared, the gamma and X-ray bands, and the survey machines that trade depth for breadth; the wide eye belongs to the survey branch and the infrared band, the heir of the ground's great sky patrols moved to the balance point, and its nearest kin in the fleet is the deep-eyed infrared successor, the two machines forming the modern pair the chapters will keep returning to, depth and width as a matched set. The reader who knows the family tree reads the machine's design as the tree's newest branch, and the chapters ahead are the branch's biography.

Chapter 1: The Discovery of Roman Space Telescope

The discovery of the wide-field space telescope is really the discovery of an idea's timing, and it begins with a gift nobody expected. The mirror at the telescope's heart was not designed for astronomy: it was built for a national reconnaissance program, an intelligence agency's optics of the same class as the famous spy-telescope hardware, and when the intelligence community concluded it had no use for a finished set of wide-field spy-optics assemblies, it passed them, in the best tradition of government serendipity, to the space agency. The gift, a mirror of the great observatories' size with a short design suited to a wide field, arrived at the exact moment the astronomy community was drafting its decadal wish list, and the astronomers, asked what they would do with such hardware, gave the answer that became the mission: a wide-field infrared surveyor, sized for the sky's biggest questions.

The second act of the discovery is the decadal process, and it deserves its paragraph because it is how modern astronomy chooses its instruments. The discipline's long-range plans, assembled by the community in ten-year

exercises, had put a wide-field infrared mission high on the list for the questions it could answer, the nature of the universe's acceleration chief among them; the gifted hardware made the mission affordable in a way a from-scratch flagship would not have been, and the plan, named at first for its surveying ambition, went through the naming, the reshaping, the descoping, and the reaffirmations that a flagship mission's childhood requires, surviving review after review until it stood confirmed as the agency's next great observatory, the survey eye to join the deep ones.

The third act is the naming, and it closes the discovery chapter with the symmetry the introduction promised. The mission, through its planning years, carried a working name from its science case; as it matured, the agency chose to name it for the astronomer whose career had made such missions possible, the first chief of the space agency's astronomy program, the organizer of the first great space telescope's earliest planning, the advocate whose office had funded the fields the mission would now survey, and the renaming made the machine a monument as well as an instrument, the first of the agency's great observatories to carry a woman's name. The telescope was thus discovered twice, once as a design, an answer to a decadal question, and once as a biography, a name that connects the wide eye to the whole history of space astronomy, and both discoveries, the idea and the name, are the chapters' foundation.

A paragraph on the earlier names the machine carried, because a flagship's aliases are its biography's marginalia. Through its planning years the mission was known by the acronym of its science case, the wide-field infrared survey telescope, a name that stated the wager plainly; the renaming to the founder's honor, announced by the agency's leadership as the mission matured through its construction reviews, was received by the field as the closing of a long-owed account, the woman's colleagues having argued for the honor across the community's meetings for years. The naming's small lesson belongs to the history: the great machines accumulate names the way careers accumulate titles, and the reader who meets the telescope under any of its aliases, the survey acronym or the founder's name, meets the same machine, the wide eye this book follows.

The gift, it should be added, was a pair. The reconnaissance program's generosity handed over two complete optical assemblies, not one, and the twin that does not fly remains one of the field's standing pieces of inventory: a finished great-observatory mirror, figured and coated and crated, resting in the agency's careful custody while its sibling becomes a surveyor. The spare has its own biography of proposals, the community's meetings having argued for years over what a second wide eye should be pointed at, a second surveyor, a different wavelength, a dedicated planet imager, the suggestions arriving each cycle the way ideas arrive at a warehouse that holds a treasure. The card for the reader is the inventory's moral: the field's real bottleneck was never the glass, it was the missions, and a finished mirror sitting in storage while the sky's questions wait is the era's quiet argument for the program-building the founder's name honors.

The mission's fifth act of discovery is the one no one enjoys and every flagship needs: the budget's gauntlet. The machine's childhood was punctuated by repeated attempts to delete it from the spending plans, the executive requests arriving with the mission absent and the legislature's answer restoring it, cycle after cycle, the community's letters and the decadal rankings doing the defending; the descopes, the cost caps, the independent reviews, were the mission's education in the realpolitik of a science budget with many claimants, and the survival is itself a discovery of sorts, the discovery that the field wanted the wide eye enough to fight for it annually. The reader should hold the fight's lesson beside the serendipity's lesson: a gifted mirror can open a door, but only a community's sustained argument walks a mission through it, and the wide eye's discovery history is the argument's, the survey wager defended until it won.